3D Simulation of Quantum Dot Devices

Application Guide

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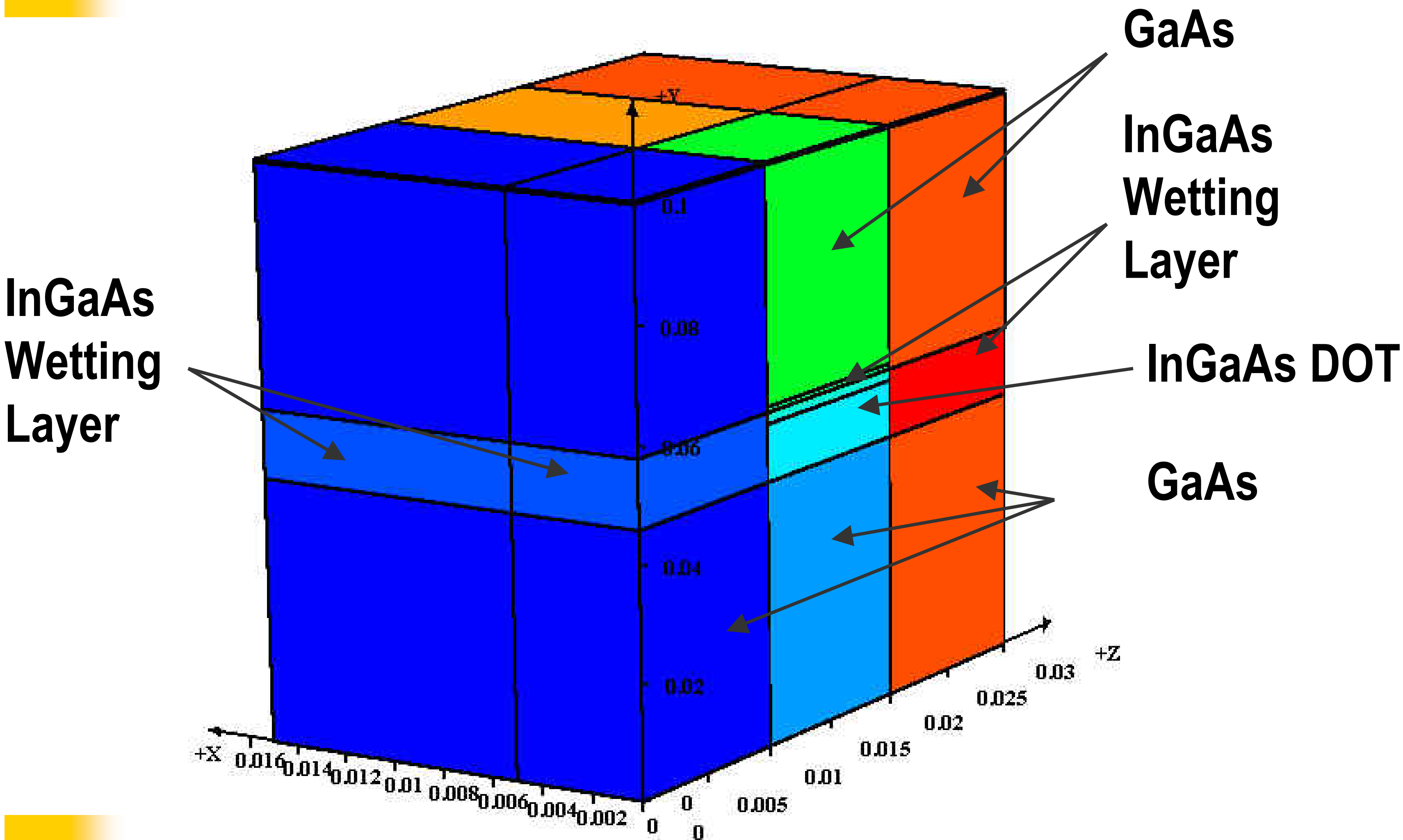
Physical Models

- Quantum dot structure offers delta-like density of states that facilitates population inversion→ Better emitters.
- Better temperature stability and lower threshold are expected.
- Modeling is separated into two parts (1) microscopic model that solves all the quantum states in a dot; (2) macroscopic model that uses (1).

Microscopic Model

- 3D structure of QDOT as set up with Layer3D/GeoEditor3D.
- Use of rectangle coordinates for QDOT shapes of boxes and pyramids, etc.
- Use of cylindrical coordinates for QDOT of columnar or cone shapes.
- Schrodinger equation is discretized in 3D and solved with sparse eigen matrix techniques.
- Strain effects are taken into account by using strain dependent potential profiles and anisotropic effective masses for different band valleys (HH,LH,CH, etc.)
- Applicable for wurtzite structure (GaN-based) as well as for zincblende structure.

A simple QDOT of a box



Use only ½ of the symmetric structure

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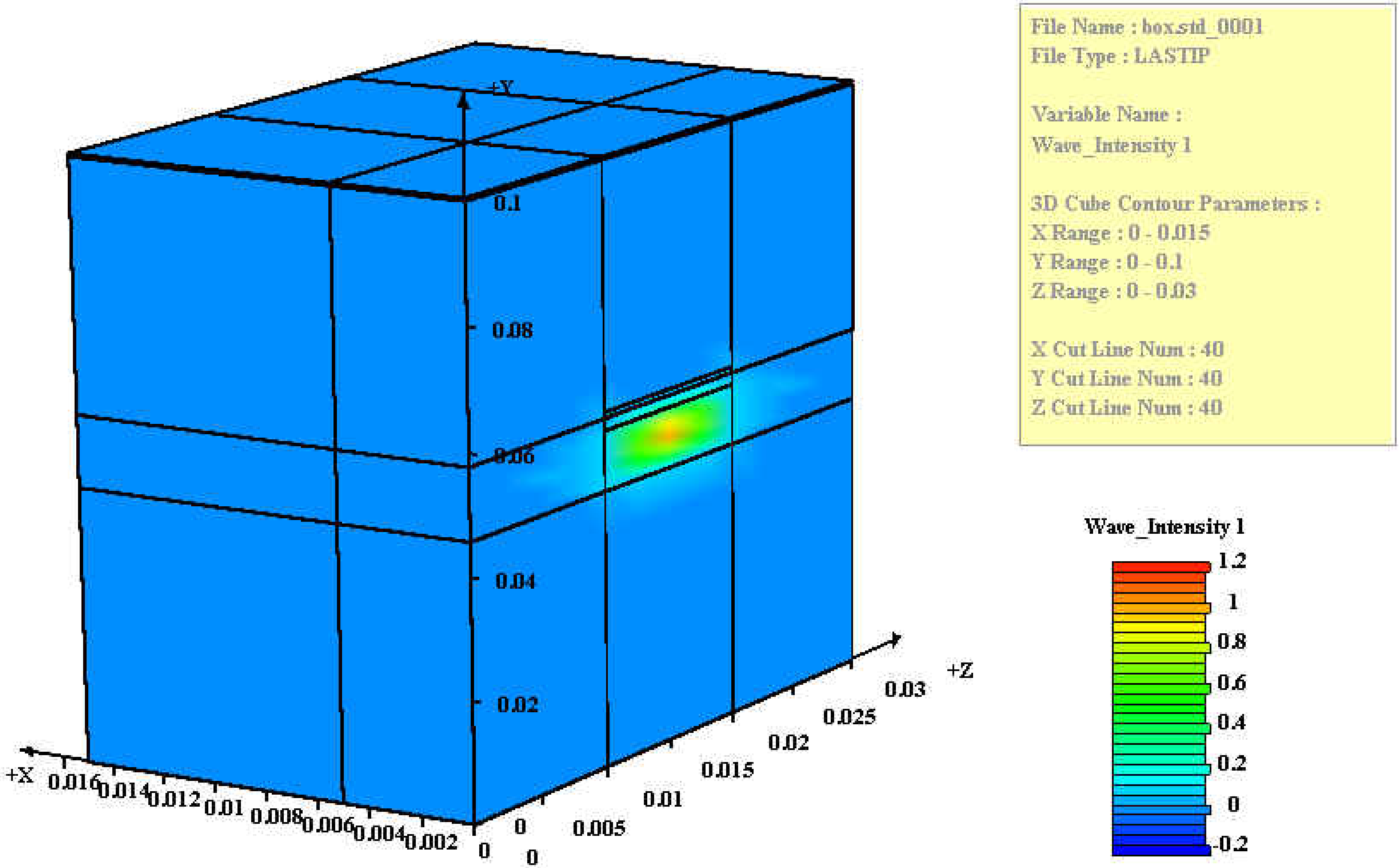
Quantum wave function of a box



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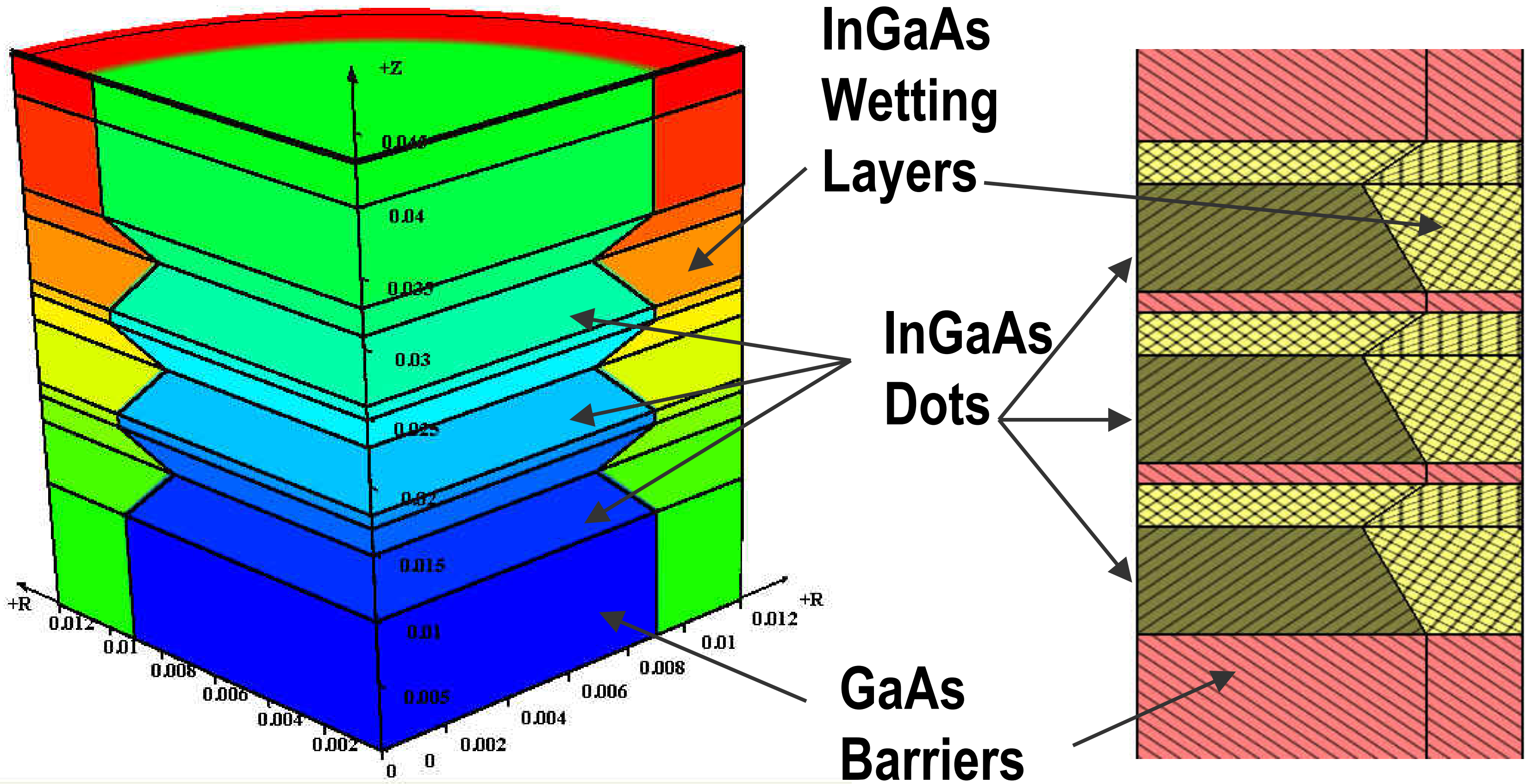
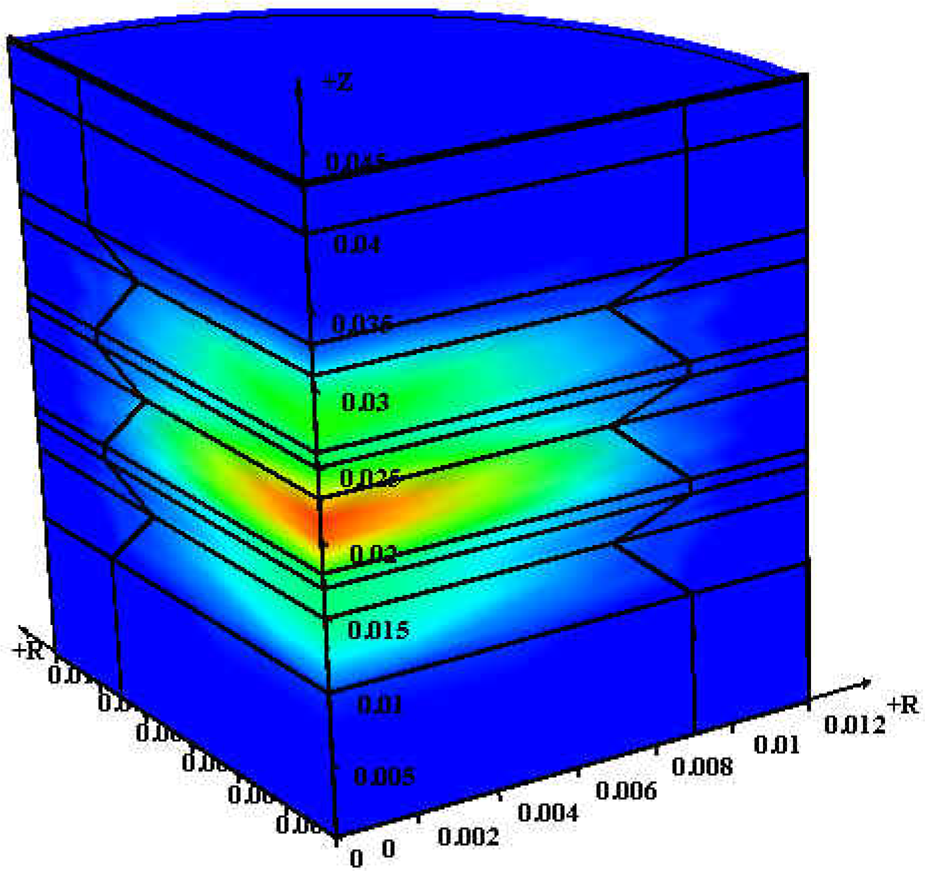
Coupled columnar dots



Remark: It is a simplified structure. Real structure
Of QDOT may have composition grading from InGaAs
To InAs at the center of the dot.

Results for columnar dots

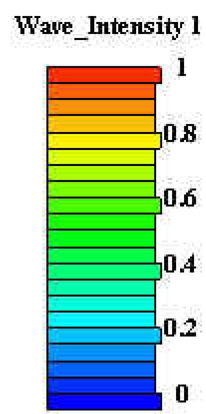


File Name : cone.std_0001
File Type : LASTIP

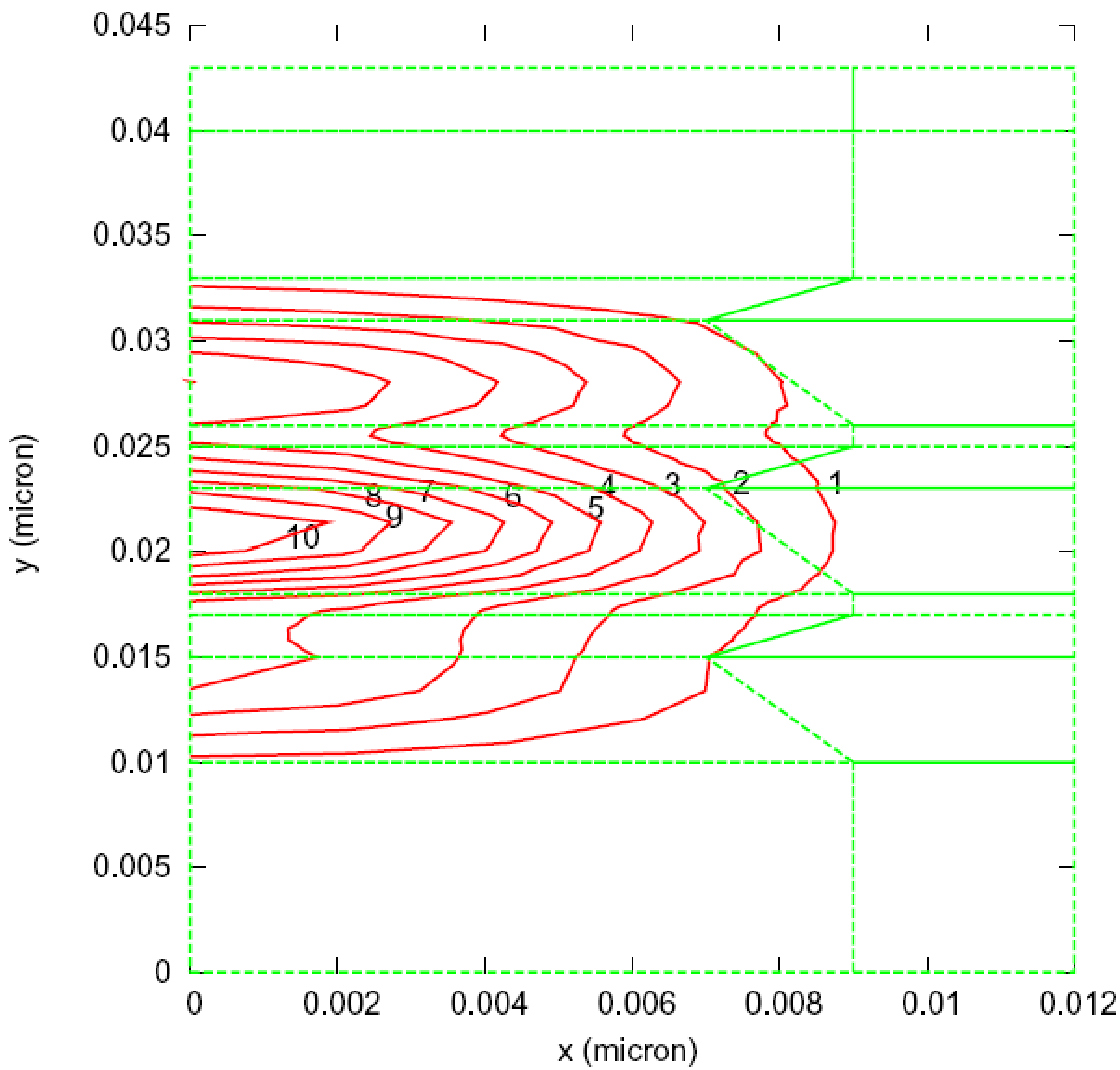
Variable Name :
Wave_Intensity 1

3D Cube Contour Parameters :
X Range : 0 - 0.012
Y Range : 0 - 0.043
Z Range : 0 - 0

X Cut Line Num : 40
Y Cut Line Num : 40
Z Cut Line Num : 40



Columnar dots (contours)



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Macroscopic Model

- Set up 2D/3D larger laser diode/LED/PD structure as usual.
- Import quantum levels and optical transition overlaps from microscopic solutions. If necessary, correct potential profile at the dot model.
- Use of delta-like density of states for optical gain, spontaneous emission and carrier concentration calculation.
- Use wetting layer as a reference to compute optical confinement and carrier sheet concentration for the dots.

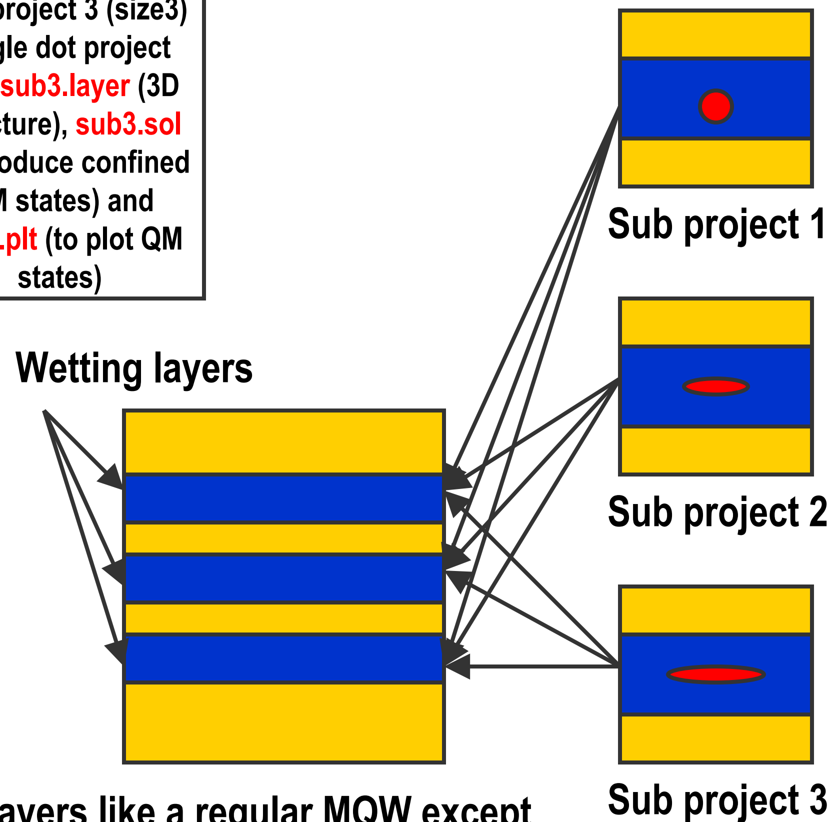
Input file system

Sub project 1 (size1) Single dot project with sub1.layer (3D structure), sub1.sol (to produce confined QM states) and sub1.plt (to plot QM states)

Sub project 2 (size2) Single dot project with sub2.layer (3D structure), sub2.sol (to produce confined QM states) and sub2.plt (to plot QM states)

Sub project 3 (size3) Single dot project with sub3.layer (3D structure), sub3.sol (to produce confined QM states) and sub3.plt (to plot QM states)

Main project with main.layer (labels MQW where QDOT will be) , main.sol (imports single QDOT results) and main.plt (to plot final results)



Main project describes wetting layers like a regular MQW except QDOT's are to be embedded within it and thus more than one material will share the wetting layers. Results of more than one single dot sub project may be imported into a main project to describe a distribution of difference sizes and shapes of dots.

Recommended Procedures

- **Set up a 3D project (or 2D under cylindrical coordinate system) to describe a single dot structure.**
- **Set up a .sol file for the 3D dot structure to produce quantum states in a .qdd file.**
- **Examine and plot the quantum states.**
- **Set up the main project .layer file to define the wetting layers where the dots will be located. Process the .layer file.**
- **Set up a .gain file (by operating on the .mater file) to import the .qdd file and plot the optical spectrum. Adjust single dot structure based on spectrum.**
- **Set up the main .sol file import the .qdd from single dot sub project and run the simulation as usual.**
- **Repeat the above steps if more sub projects are required.**

Sub Project .sol File

```
cylindrical

wave_boundary point_ll=(0. 0.) point_ur=(0.012 0.043)
init_wave init_wavel=1.3
multimode mode_num=20 boundary_type1=(2 1 1 1) boundary_type2=(1 1 1 1)
sparse_eigen_solver use_mf=yes
direct_eigen

solve_lateral_wave quantum_wave=yes cond_valley=1 val_valley=1 &&
 export_data=cone.qdd
solve_lateral_wave quantum_wave=yes cond_valley=1 val_valley=2
solve_lateral_wave quantum_wave=yes cond_valley=2 val_valley=1
```

- **Set up the .sol as if you are solving for multiple lateral optical modes (see multimode above) in a wave guide. Only difference is that it is Schrodinger equation in 3D.**
- **Use direct_eigen and use_mf to solve the eigen modes.**
- **solve_lateral_wave is the main task, output in a .qdd file.**

Main Project .layer File

```
$ --- wetting layer
layer_mater  macro_name=ingaas &&
  var_symbol1=x var1=0.250 &&
  active_macro=InGaAs/AlGaAs &&
  avar_symbol1=xw avar1=0.250 &&
  avar_symbol2=xb avar2=  0.000000E+00 &&
  column_num=1

qdot_layer_mater macro_name=ingaas active_macro=cx-InGaAs &&
  var_symbol1=x avar_symbol1=xw var1=0.480 avar1=0.480 &&
  column_num=1 height=0.01

layer d=0.006 n=  3 &&
  r=  0.120000E+01
$ --- end wetting layer
```

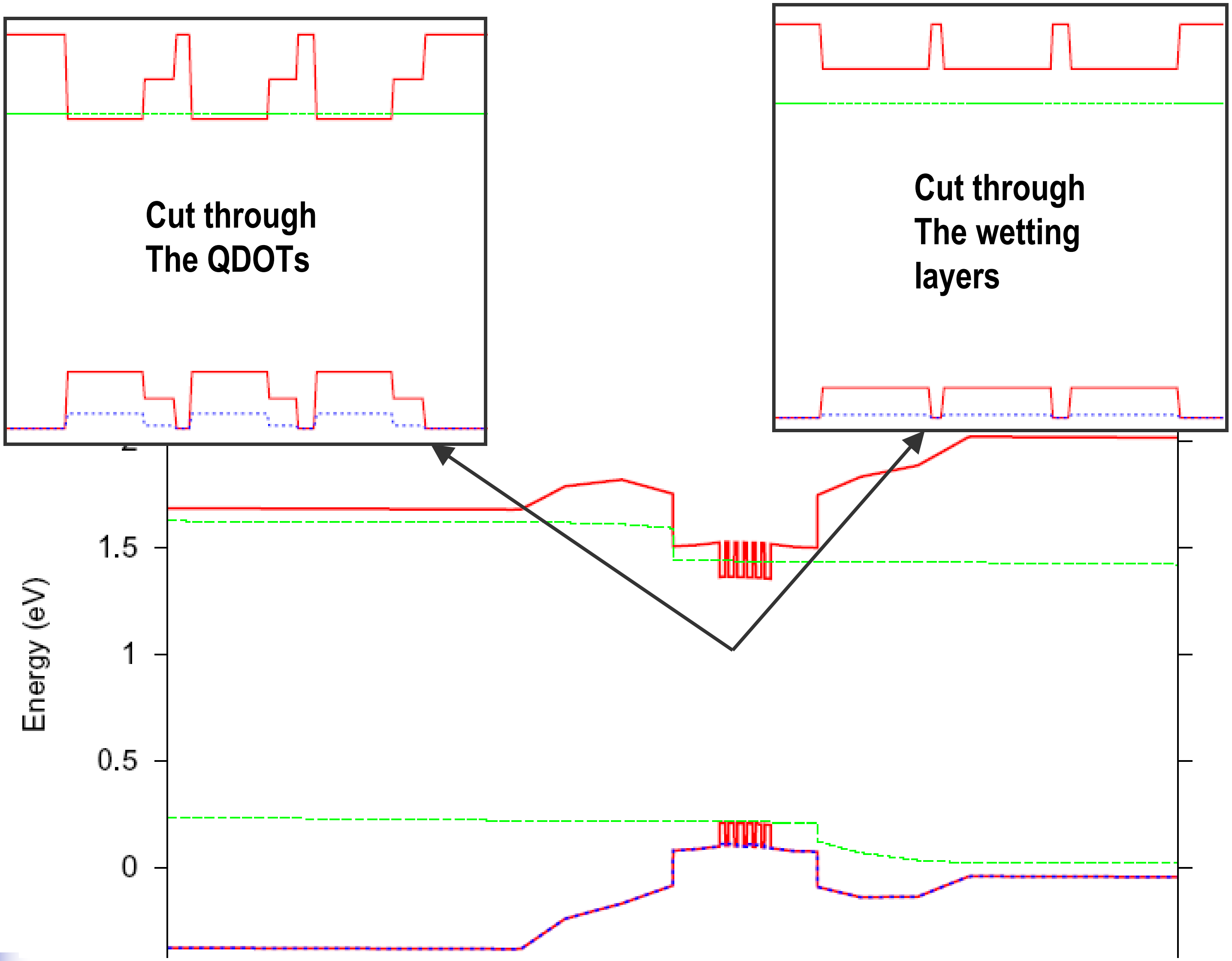
- **Use layer_mater to define the wetting layer where the dots will be embedded.**
- **Use qdot_layer_mater immediately after to define one of the dots to be embedded. No need to specify more than one since the importation will take care of all dots.**

Main Project .sol File

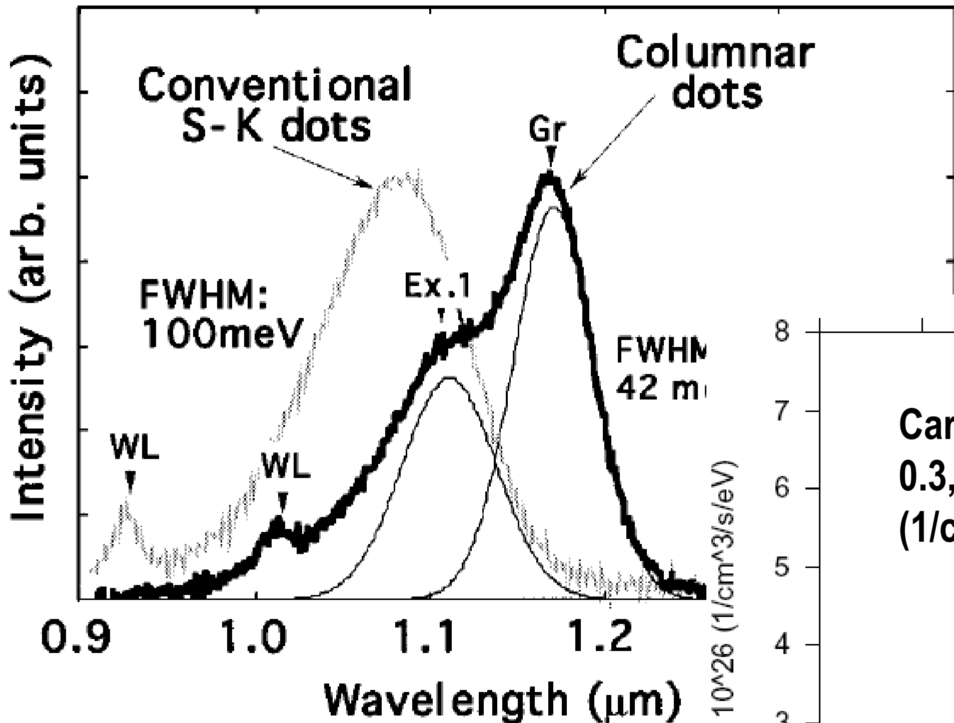
```
qdot_material surface_density=1.e15 tau_broaden=0.5e-13 &&
 mater_active=4 import_dir=cone2 import_qdot_data=cone.qdd &&
 coupled_dots=3
```

- **The main .sol file just need to import the sub project using qdot_material statement.**
- **One also needs to set dot density and gain spectrum broadening.**
- **The simulator will take care of all the rest.**

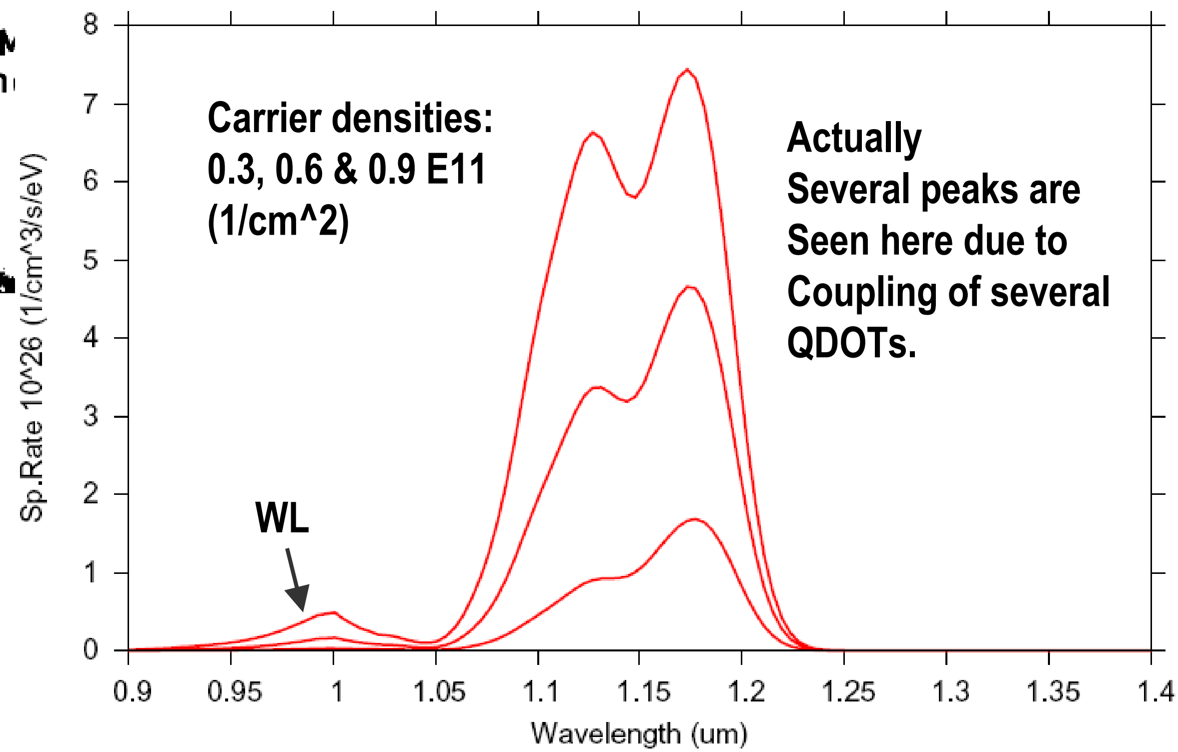
Band diagrams



Comparing PL with experiment

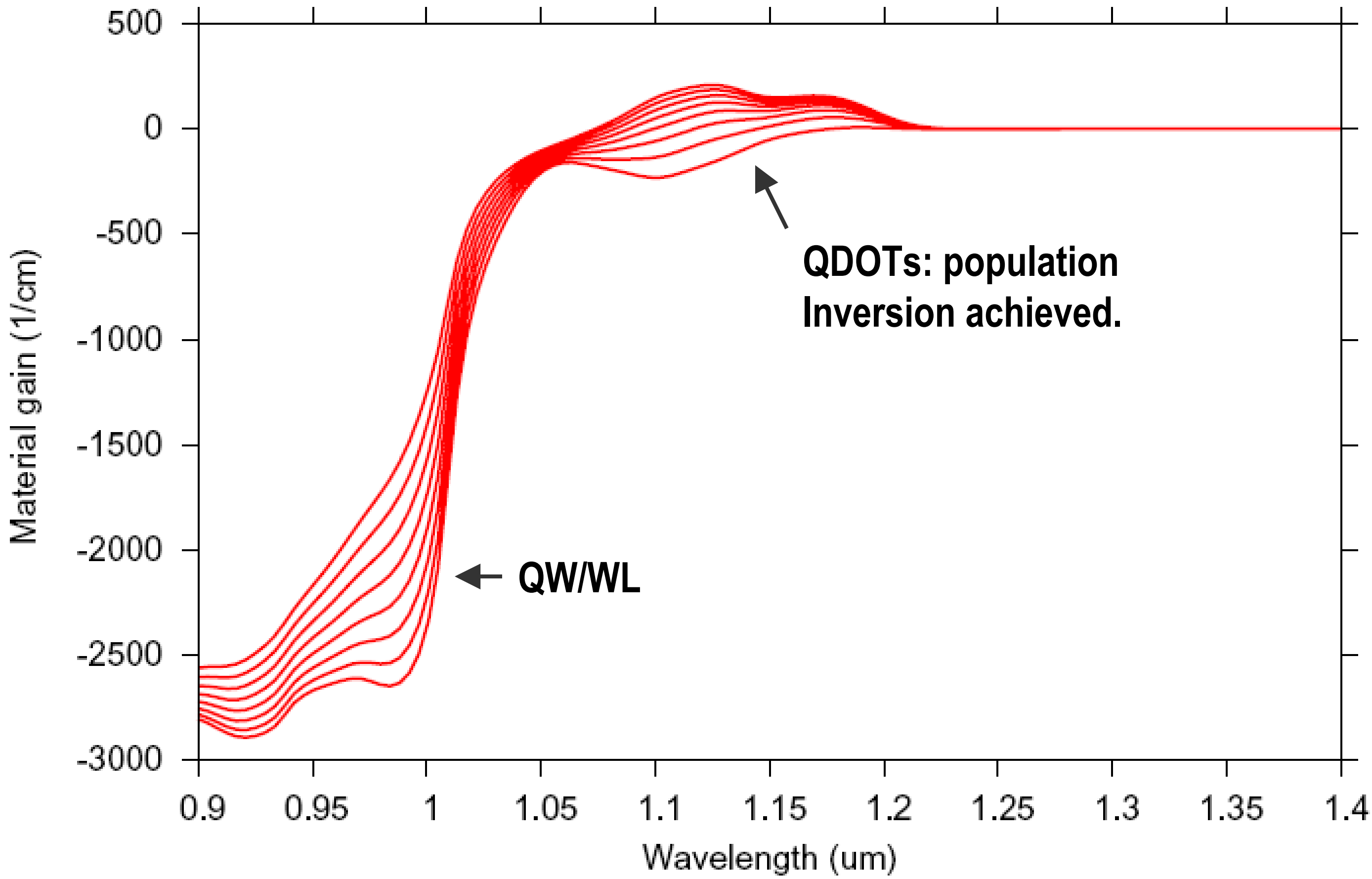


M. Sugawara, et. al.,
IEEE J. Select. Topics in
Quant. Electron. vol. 6,p462
2000.



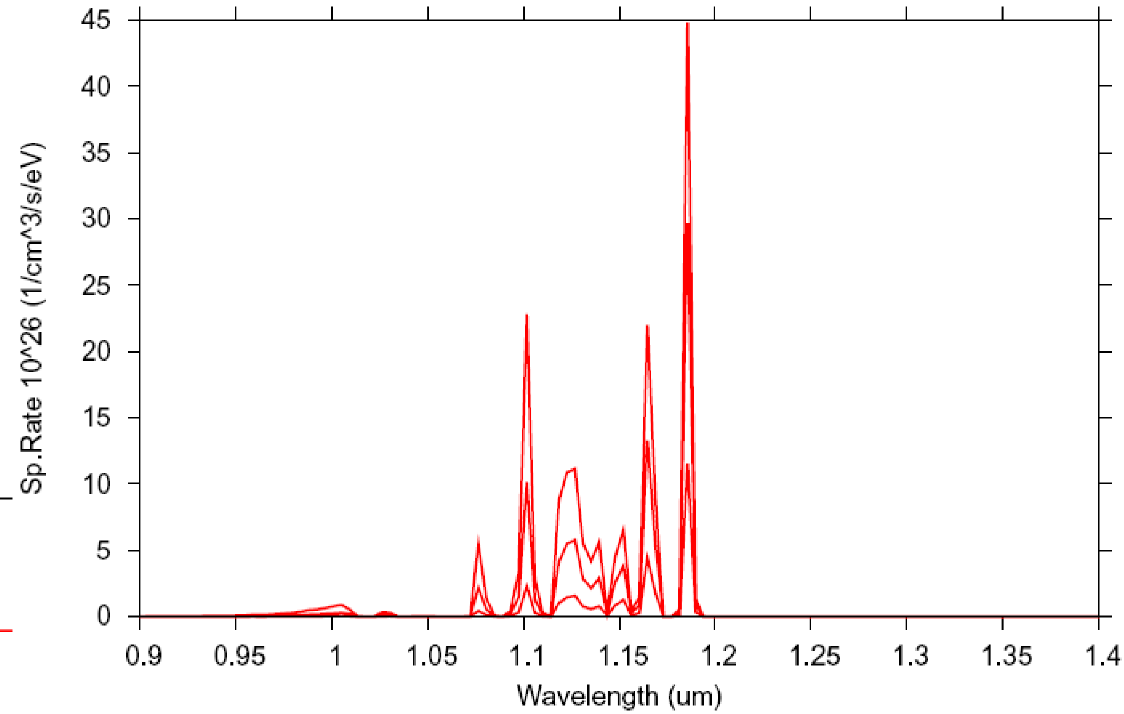
Both QW and QDOT models are needed here.

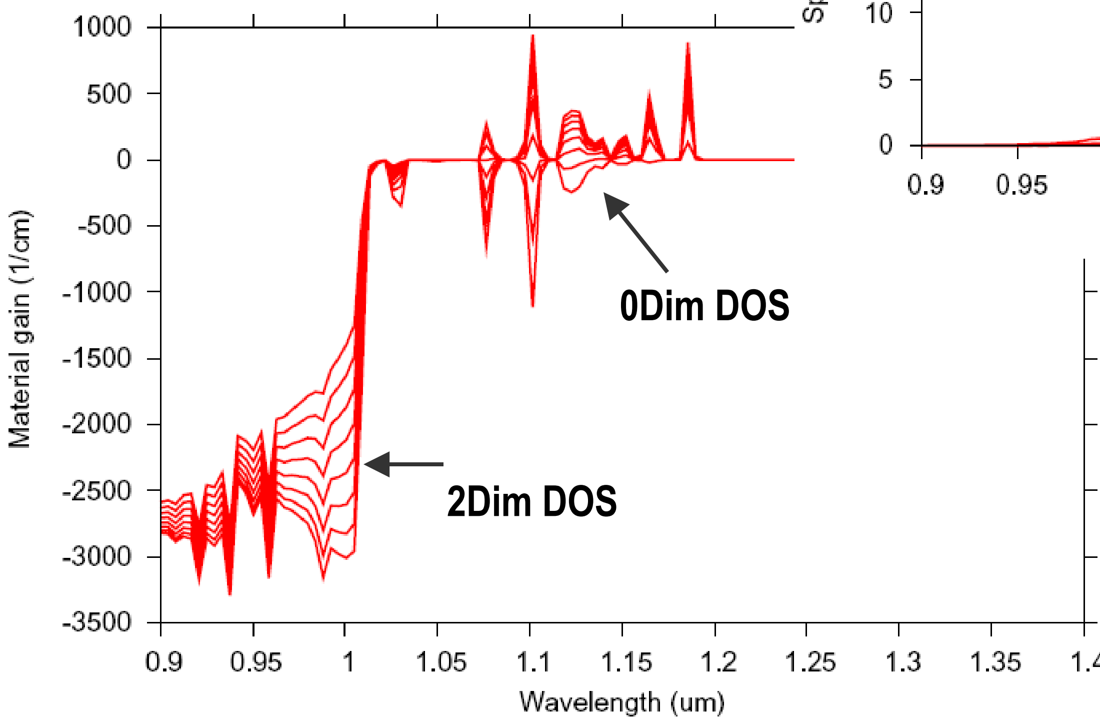
Optical gain spectrum



Carrier conc. in wetting layers ranges from 0.6E11 ➔ 6E11 (1/cm^2). For QDOT LED, this will be useful for photon-recycling model.

Spectrum without broadening



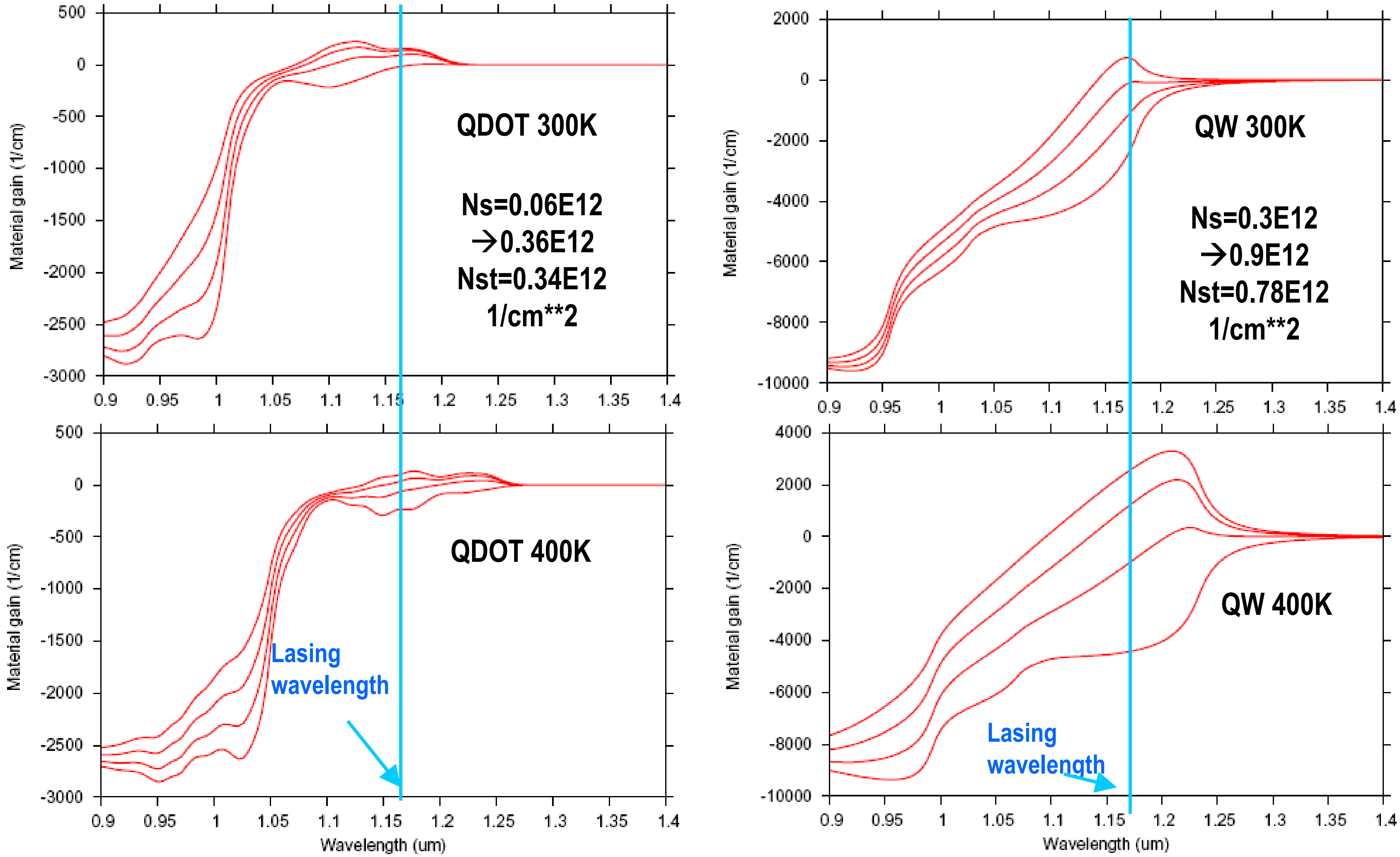


Same conditions as previous Pages.

Temperature dependence study

- **Objective: to compare performance of MQW and QDOT lasers at the same wavelength (1.17 microns) using the same material system (InGaAs/AlGaAs).**
- **Both QDOT and QW indium compositions are tuned to have gain peak at 1.17 microns at T=300K.**
- **Assumption of no current spreading (1D) simulation, no temperature dependence in non-radiative carrier lifetime and Auger recombination.**

Gain Spectra



All four simulations are forced to lase at the same wavelength.
Ns is the carrier surface density and Nst is threshold surface density.
Remark: QDOT spectra less sensitive to temperature change.

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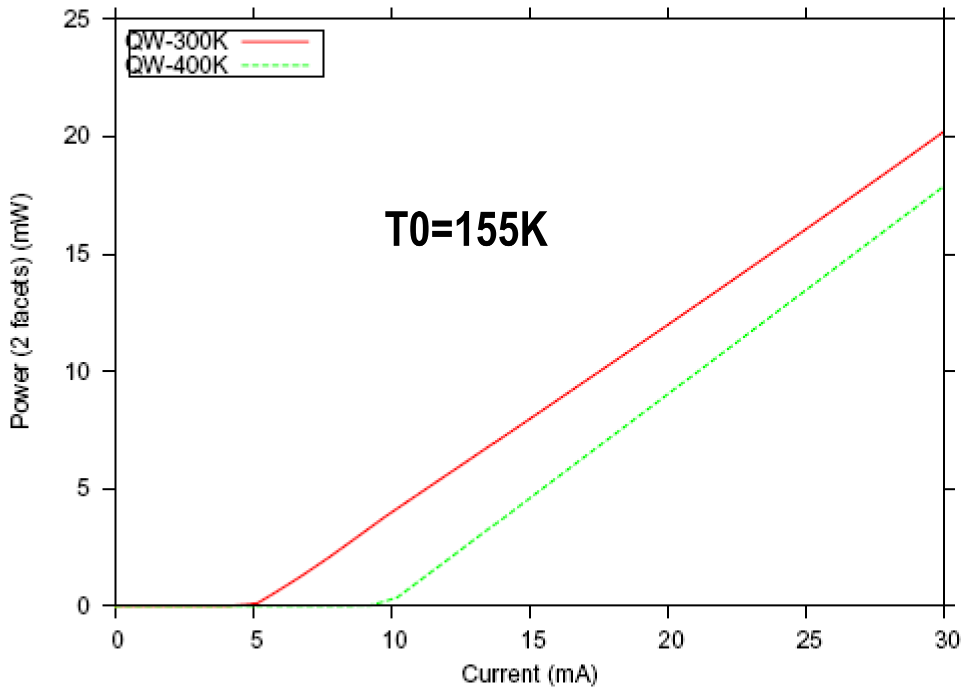
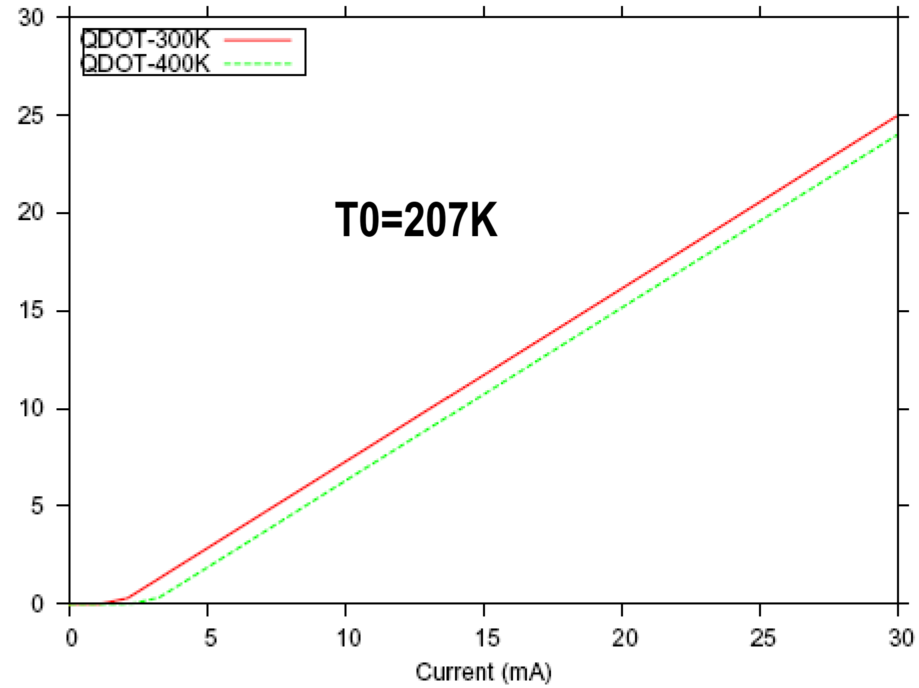
Lasing behaviour





Comparison of lasing behavior at 300 and 400 K. All structures are forced to lase at 1.17 microns and temperature dependences in bandgap, optical gain, carrier concentration, and radiative recombination have been taken into account. All other quantities are assumed to be temperature independent for simplicity

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Summary

- An efficient QDOT model has been established for PICS3D/LASTIP/APSYS.
- QDOT model may be used for photo-detectors as well as for laser diodes/LED since gain/absorption model is implemented in the same way as QW model.
- Combination of microscopic and macroscopic modeling of QDOT offers both efficiency and accuracy.
- Simulated lasing behaviors for LD indicate substantial benefits for lasing threshold and temperature insensitivity.